```
<!DOCTYPE html>
<html lang="fr">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Smart Assistance Manager</title>
  <!-- Tailwind CSS -->
  <script src="https://cdn.tailwindcss.com"></script>
  <!-- FontAwesome -->
  <link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
  <!-- Firebase Compat CDN (Pour la synchronisation collaborative en direct) -->
  <script src="https://www.gstatic.com/firebasejs/9.22.0/firebase-app-compat.js"></script>
  <script src="https://www.gstatic.com/firebasejs/9.22.0/firebase-database-compat.js"></script>
  <style>
    @keyframes pulse-subtle {
      0%, 100% { transform: scale(1); }
      50% { transform: scale(1.02); }
    }
    .animate-pulse-btn {
      animation: pulse-subtle 2s infinite ease-in-out;
    }
  </style>
</head>
<body class="bg-slate-100 text-slate-800 min-h-screen font-sans antialiased flex flex-col">

  <!-- ================= TOPBAR ================= -->
  <header class="bg-indigo-950 text-white shadow-lg sticky top-0 z-40 border-b
border-indigo-900">
    <div class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 h-16 flex items-center justify-between">
      <div class="flex items-center space-x-3">
        <div class="bg-indigo-600 p-2 rounded-xl text-white shadow">
          <i class="fa-solid fa-headset text-xl"></i>
        </div>
        <div>
          <span class="text-xl font-black tracking-tight">HelpDesk <span
class="text-indigo-400">Live</span></span>
          <span id="roleIndicator" class="text-xs text-indigo-300 block font-medium">Portail
d'Assistance</span>
        </div>
      </div>

      <div class="flex items-center space-x-3" id="headerUserNav"></div>
    </div>
```

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  </header>

  <!-- ================= CONTENEUR PRINCIPAL ================= -->
  <main class="max-w-7xl mx-auto px-4 sm:px-6 lg:px-8 py-6 flex-1 w-full">

    <!-- ================= 1. ACCUEIL : SÉLECTION D'ACCÈS ================= -->
    <div id="viewLoginSelector" class="max-w-xl mx-auto mt-6 space-y-6">
      <div class="text-center space-y-2">
        <h1 class="text-3xl font-extrabold text-slate-900">Assistance Opérationnelle</h1>
        <p class="text-sm text-slate-500">Sélectionnez votre profil pour accéder à votre espace de
travail.</p>
      </div>

      <div class="grid grid-cols-1 sm:grid-cols-2 gap-4">
        <!-- Connexion Opérationnelle (Agent & Support) -->
        <div class="bg-white p-6 rounded-2xl shadow-sm border border-slate-200
hover:border-indigo-500 transition flex flex-col justify-between space-y-4">
          <div>
            <div class="w-12 h-12 bg-indigo-50 text-indigo-600 rounded-xl flex items-center
justify-center text-xl mb-3">
              <i class="fa-solid fa-users"></i>
            </div>
            <h3 class="font-bold text-slate-800 text-lg">Agents & Support QA</h3>
            <p class="text-xs text-slate-500 mt-1">Connexion instantanée sans mot de passe :
saisissez uniquement votre nom et activité.</p>
          </div>
          <button onclick="showOperativeLogin()" class="w-full py-2.5 bg-indigo-600
hover:bg-indigo-700 text-white font-bold rounded-xl text-sm transition">
            Entrer (Agent / QA)
          </button>
        </div>

        <!-- Accès Administrateur Sécurisé -->
        <div class="bg-white p-6 rounded-2xl shadow-sm border border-slate-200
hover:border-amber-500 transition flex flex-col justify-between space-y-4">
          <div>
            <div class="w-12 h-12 bg-amber-50 text-amber-600 rounded-xl flex items-center
justify-center text-xl mb-3">
              <i class="fa-solid fa-shield-halved"></i>
            </div>
            <h3 class="font-bold text-slate-800 text-lg">Espace Administration</h3>
            <p class="text-xs text-slate-500 mt-1">Accès protégé par mot de passe. Gestion des
menus, listes et création d'admins.</p>
          </div>
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        <button onclick="showAdminLogin()" class="w-full py-2.5 bg-slate-900 hover:bg-black
text-amber-300 font-bold rounded-xl text-sm transition">
          Connexion Admin <i class="fa-solid fa-lock text-xs ml-1"></i>
        </button>
      </div>
    </div>

    <!-- Barre de partage rapide du lien -->
    <div class="bg-white p-4 rounded-xl border border-slate-200 flex flex-col sm:flex-row
items-center justify-between gap-3 shadow-sm">
      <div class="flex items-center gap-2">
        <i class="fa-solid fa-share-nodes text-indigo-600"></i>
        <span class="text-xs text-slate-700 font-medium">Partagez ce site avec vos collègues
:</span>
      </div>
      <button onclick="copyCurrentUrl()" class="px-4 py-2 bg-indigo-50 hover:bg-indigo-100
text-indigo-700 text-xs font-bold rounded-xl transition flex items-center gap-1.5">
        <i class="fa-solid fa-copy"></i> Copier l'URL du site
      </button>
    </div>
  </div>

  <!-- ================= 2. FORMULAIRE AGENT / SUPPORT ================= -->
  <div id="viewOperativeAuth" class="max-w-md mx-auto mt-6 bg-white p-8 rounded-2xl
shadow-md border border-slate-200 hidden">
    <div class="flex items-center justify-between mb-5">
      <h2 class="text-xl font-bold text-slate-800">Accès Opérationnel</h2>
      <button onclick="showView('loginSelector')" class="text-slate-400 hover:text-slate-600
text-sm">
        <i class="fa-solid fa-arrow-left"></i> Retour
      </button>
    </div>

    <form id="formOperativeAuth" class="space-y-4">
      <div>
        <label class="block text-xs font-bold uppercase tracking-wider text-slate-600 mb-1">Votre
Nom & Prénom</label>
        <input type="text" id="opName" required placeholder="Ex: Julie Martin" class="w-full px-4
py-2.5 rounded-xl border border-slate-300 focus:ring-2 focus:ring-indigo-500 focus:outline-none
text-sm">
      </div>

      <div>
```

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        <label class="block text-xs font-bold uppercase tracking-wider text-slate-600
mb-1">Activité / Projet</label>
        <select id="opActivity" required class="w-full px-4 py-2.5 rounded-xl border
border-slate-300 focus:ring-2 focus:ring-indigo-500 focus:outline-none text-sm
bg-white"></select>
       </div>

       <div>
        <label class="block text-xs font-bold uppercase tracking-wider text-slate-600 mb-1">Votre
Rôle</label>
        <div class="grid grid-cols-2 gap-3">
         <label class="border-2 border-slate-200 rounded-xl p-3 flex items-center space-x-2
cursor-pointer hover:bg-slate-50 transition">
           <input type="radio" name="opRole" value="agent" checked class="text-indigo-600">
           <span class="text-sm font-semibold text-slate-700"><i class="fa-solid fa-user
text-slate-400 mr-1"></i> Agent</span>
         </label>
         <label class="border-2 border-slate-200 rounded-xl p-3 flex items-center space-x-2
cursor-pointer hover:bg-slate-50 transition">
           <input type="radio" name="opRole" value="support" class="text-indigo-600">
           <span class="text-sm font-semibold text-slate-700"><i class="fa-solid
fa-chalkboard-user text-indigo-500 mr-1"></i> QA / Trainer</span>
         </label>
        </div>
       </div>

       <button type="submit" class="w-full py-3 bg-indigo-600 hover:bg-indigo-700 text-white
font-bold rounded-xl shadow-md transition">
        Valider et Entrer
       </button>
     </form>
   </div>

   <!-- ================= 3. FORMULAIRE ADMIN (MOT DE PASSE)
================= -->
   <div id="viewAdminAuth" class="max-w-md mx-auto mt-6 bg-white p-8 rounded-2xl
shadow-xl border border-slate-200 hidden">
     <div class="flex items-center justify-between mb-5">
       <h2 class="text-xl font-bold text-slate-800 flex items-center gap-2">
         <i class="fa-solid fa-lock text-amber-500"></i> Accès Administrateur
       </h2>
       <button onclick="showView('loginSelector')" class="text-slate-400 hover:text-slate-600
text-sm">
         <i class="fa-solid fa-arrow-left"></i> Retour
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      </button>
    </div>

    <div id="adminFirstSetupNotice" class="hidden mb-4 p-3 bg-amber-50 border
border-amber-200 rounded-xl text-xs text-amber-800">
      <i class="fa-solid fa-info-circle mr-1"></i> <strong>Premier lancement :</strong> Créez le
premier compte Administrateur de la plateforme.
    </div>

    <form id="formAdminAuth" class="space-y-4">
      <div>
        <label class="block text-xs font-bold uppercase tracking-wider text-slate-600
mb-1">Identifiant Admin</label>
        <input type="text" id="adminAuthName" required placeholder="Ex: admin" class="w-full
px-4 py-2.5 rounded-xl border border-slate-300 focus:ring-2 focus:ring-amber-500 text-sm">
      </div>

      <div>
        <label class="block text-xs font-bold uppercase tracking-wider text-slate-600 mb-1">Mot
de passe</label>
        <input type="password" id="adminAuthPassword" required placeholder="••••••••"
class="w-full px-4 py-2.5 rounded-xl border border-slate-300 focus:ring-2 focus:ring-amber-500
text-sm">
      </div>

      <button type="submit" id="btnAdminSubmit" class="w-full py-3 bg-slate-900 hover:bg-black
text-amber-300 font-bold rounded-xl shadow transition">
        Se connecter
      </button>
    </form>
  </div>

  <!-- ================= 4. ESPACE DE TRAVAIL CONNECTÉ ================= -->
  <div id="viewWorkspace" class="space-y-6 hidden">

    <!-- Barre d'onglets (filtrée selon les droits du rôle) -->
    <div class="flex items-center justify-between border-b border-slate-200 pb-3">
      <nav class="flex space-x-2 sm:space-x-3" id="navigationTabs">
        <!-- Onglet Agent -->
        <button onclick="switchTab('agent')" id="tabBtnAgent" class="px-4 py-2 rounded-xl
font-bold text-xs sm:text-sm flex items-center gap-2">
          <i class="fa-solid fa-circle-question"></i> Mon Assistance
        </button>
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        <!-- Onglet File Globale (Support & Admin) -->
        <button onclick="switchTab('support')" id="tabBtnSupport" class="px-4 py-2 rounded-xl
font-bold text-xs sm:text-sm flex items-center gap-2 hidden">
          <i class="fa-solid fa-list-check"></i> File d'Assistance
          <span id="badgePendingCount" class="bg-rose-500 text-white text-xs px-2 py-0.5
rounded-full font-black">0</span>
        </button>

        <!-- Onglet Statistiques (Support & Admin) -->
        <button onclick="switchTab('stats')" id="tabBtnStats" class="px-4 py-2 rounded-xl
font-bold text-xs sm:text-sm flex items-center gap-2 hidden">
          <i class="fa-solid fa-chart-pie"></i> Statistiques & Données
        </button>

        <!-- Onglet Configuration (Admin uniquement) -->
        <button onclick="switchTab('admin')" id="tabBtnAdmin" class="px-4 py-2 rounded-xl
font-bold text-xs sm:text-sm flex items-center gap-2 hidden">
          <i class="fa-solid fa-sliders"></i> Administration
        </button>
      </nav>

      <div id="syncBadge" class="flex items-center gap-2 text-xs bg-white px-3 py-1.5
rounded-xl border shadow-sm">
        <span id="syncDot" class="w-2 h-2 rounded-full bg-slate-400"></span>
        <span id="syncText" class="font-medium text-slate-600">Local</span>
      </div>
    </div>

    <!-- ================= SECTION AGENT ================= -->
    <section id="tabAgent" class="space-y-6">
      <div class="bg-gradient-to-r from-indigo-950 via-indigo-900 to-indigo-950 rounded-3xl p-6
sm:p-8 text-white shadow-xl flex flex-col md:flex-row items-center justify-between gap-6">
        <div class="space-y-2">
          <span class="inline-flex items-center px-3 py-1 rounded-full text-xs font-semibold
bg-indigo-500/30 text-indigo-200">
            <span class="w-2 h-2 rounded-full bg-emerald-400 mr-2 animate-ping"></span>
Équipe Support Disponible
          </span>
          <h2 class="text-2xl sm:text-3xl font-extrabold tracking-tight">Un blocage sur un
document ?</h2>
          <p class="text-indigo-200 text-sm max-w-xl">
            Cliquez ci-contre pour transmettre votre interrogation. Choisissez si votre question
concerne la Nature, la Date ou le Numéro du document.
          </p>
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        </div>
        <button onclick="openRequestModal()" class="animate-pulse-btn whitespace-nowrap
px-8 py-4 bg-rose-500 hover:bg-rose-600 text-white text-base sm:text-lg font-black rounded-2xl
shadow-2xl flex items-center gap-3 transition">
          <i class="fa-solid fa-triangle-exclamation text-2xl"></i> Demander de l'aide
        </button>
      </div>

      <!-- Demandes personnelles de l'agent -->
      <div class="bg-white rounded-2xl shadow-sm border border-slate-200 overflow-hidden">
        <div class="px-6 py-4 border-b border-slate-100 flex items-center justify-between">
          <h3 class="font-bold text-slate-800">Mes demandes récentes</h3>
          <span class="text-xs text-slate-400">Vos demandes uniquement (cloisonné)</span>
        </div>
        <div class="overflow-x-auto">
          <table class="w-full text-left text-sm">
            <thead class="bg-slate-50 text-slate-500 uppercase text-xs">
              <tr>
                <th class="px-6 py-3">Heure</th>
                <th class="px-6 py-3">Document</th>
                <th class="px-6 py-3">Type & Sous-type</th>
                <th class="px-6 py-3">Question sur</th>
                <th class="px-6 py-3">Statut</th>
                <th class="px-6 py-3">Pris en charge par</th>
                <th class="px-6 py-3">Action</th>
              </tr>
            </thead>
            <tbody id="agentTicketsTable" class="divide-y divide-slate-100"></tbody>
          </table>
        </div>
      </div>
    </section>

    <!-- ================= SECTION SUPPORT / QA ================= -->
    <section id="tabSupport" class="space-y-6 hidden">
      <div class="flex flex-wrap items-center justify-between gap-4">
        <div>
          <h2 class="text-xl font-bold text-slate-900">File d'Assistance en Direct</h2>
          <p class="text-sm text-slate-500">Toutes les sollicitations du plateau. Alertes sonores
activées.</p>
        </div>
        <div class="flex items-center gap-3">
          <button onclick="testSoundNotification()" class="px-3 py-2 text-xs bg-slate-200
hover:bg-slate-300 font-bold rounded-xl text-slate-700">
```

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            <i class="fa-solid fa-volume-high mr-1"></i> Tester le son
          </button>
          <select id="filterActivity" onchange="renderSupportTickets()" class="text-xs border
rounded-xl px-3 py-2 bg-white font-semibold">
            <option value="ALL">Toutes les activités</option>
          </select>
        </div>
      </div>

      <div id="supportTicketsContainer" class="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3
gap-5"></div>
    </section>

    <!-- ================= SECTION STATISTIQUES ================= -->
    <section id="tabStats" class="space-y-6 hidden">
      <div>
        <h2 class="text-xl font-bold text-slate-900">Statistiques & Analyse</h2>
        <p class="text-sm text-slate-500">Volumes globaux, suivi par QA/Trainer et recours par
agent.</p>
      </div>

      <div class="grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-4 gap-4">
        <div class="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm flex
items-center justify-between">
          <div>
            <p class="text-xs uppercase text-slate-400 font-bold">Total Demandes</p>
            <h3 class="text-3xl font-black text-slate-800 mt-1" id="statTotalRequests">0</h3>
          </div>
          <div class="w-12 h-12 bg-indigo-50 text-indigo-600 rounded-xl flex items-center
justify-center text-xl">
            <i class="fa-solid fa-folder-open"></i>
          </div>
        </div>

        <div class="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm flex
items-center justify-between">
          <div>
            <p class="text-xs uppercase text-slate-400 font-bold">En Attente</p>
            <h3 class="text-3xl font-black text-amber-500 mt-1" id="statPendingRequests">0</h3>
          </div>
          <div class="w-12 h-12 bg-amber-50 text-amber-600 rounded-xl flex items-center
justify-center text-xl">
            <i class="fa-solid fa-hourglass-half"></i>
          </div>
```

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        </div>

        <div class="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm flex
items-center justify-between">
          <div>
            <p class="text-xs uppercase text-slate-400 font-bold">En Cours</p>
            <h3 class="text-3xl font-black text-blue-600 mt-1" id="statProgressRequests">0</h3>
          </div>
          <div class="w-12 h-12 bg-blue-50 text-blue-600 rounded-xl flex items-center
justify-center text-xl">
            <i class="fa-solid fa-spinner"></i>
          </div>
        </div>

        <div class="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm flex
items-center justify-between">
          <div>
            <p class="text-xs uppercase text-slate-400 font-bold">Résolues</p>
            <h3 class="text-3xl font-black text-emerald-600 mt-1"
id="statResolvedRequests">0</h3>
          </div>
          <div class="w-12 h-12 bg-emerald-50 text-emerald-600 rounded-xl flex items-center
justify-center text-xl">
            <i class="fa-solid fa-circle-check"></i>
          </div>
        </div>
      </div>

      <div class="grid grid-cols-1 lg:grid-cols-2 gap-6">
        <div class="bg-white rounded-2xl shadow-sm border border-slate-200 p-5">
          <h3 class="font-bold text-slate-800 mb-4 flex items-center gap-2">
            <i class="fa-solid fa-chalkboard-user text-indigo-600"></i> Traités par QA & Trainer
          </h3>
          <div class="overflow-x-auto">
            <table class="w-full text-left text-sm">
              <thead class="bg-slate-50 text-slate-500 uppercase text-xs">
                <tr>
                  <th class="px-4 py-2.5">QA / Trainer</th>
                  <th class="px-4 py-2.5">En cours</th>
                  <th class="px-4 py-2.5">Résolus</th>
                  <th class="px-4 py-2.5 text-right">Total</th>
                </tr>
              </thead>
              <tbody id="statsQaTable" class="divide-y divide-slate-100"></tbody>
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          </table>
        </div>
      </div>

      <div class="bg-white rounded-2xl shadow-sm border border-slate-200 p-5">
        <h3 class="font-bold text-slate-800 mb-4 flex items-center gap-2">
          <i class="fa-solid fa-users text-indigo-600"></i> Demandes par Agent
        </h3>
        <div class="overflow-x-auto">
          <table class="w-full text-left text-sm">
            <thead class="bg-slate-50 text-slate-500 uppercase text-xs">
              <tr>
                <th class="px-4 py-2.5">Agent</th>
                <th class="px-4 py-2.5">Activité</th>
                <th class="px-4 py-2.5 text-right">Total</th>
              </tr>
            </thead>
            <tbody id="statsAgentTable" class="divide-y divide-slate-100"></tbody>
          </table>
        </div>
      </div>
    </div>
  </section>

  <!-- ================= SECTION ADMIN ================= -->
  <section id="tabAdmin" class="space-y-6 hidden">
    <div>
      <h2 class="text-xl font-bold text-slate-900">Console d'Administration</h2>
      <p class="text-sm text-slate-500">Gérez les administrateurs, paramétrez les listes et
configurez la synchronisation en ligne.</p>
    </div>

    <!-- Synchronisation Cloud (Firebase) pour multi-postes -->
    <div class="bg-white p-6 rounded-2xl border-2 border-indigo-200 shadow-sm space-y-3">
      <div class="flex items-center justify-between">
        <h3 class="font-bold text-slate-800 text-base flex items-center gap-2">
          <i class="fa-solid fa-cloud text-indigo-600"></i> Synchronisation Multi-postes (Firebase
Gratuit)
        </h3>
        <span id="adminSyncBadge" class="text-xs font-bold px-2.5 py-1 rounded-full
bg-slate-100 text-slate-600">Non configuré</span>
      </div>
      <p class="text-xs text-slate-500">
```

Pour que tous les collègues connectés sur des ordinateurs différents voient les alertes en direct, collez votre objet de configuration Firebase (ou l'URL de votre base Realtime Database).
</p>

<input type="text" id="firebaseConfigInput" placeholder='https://mon-helpdesk-default-rtdb.firebaseio.com/' class="flex-1 px-3 py-2 border rounded-xl text-xs bg-slate-50 font-mono">
<button onclick="saveFirebaseConfig()" class="px-4 py-2 bg-indigo-600 hover:bg-indigo-700 text-white font-bold rounded-xl text-xs">
Connecter le Cloud
</button>

<!-- Gestion des Administrateurs -->

<h3 class="font-bold text-slate-800 text-base flex items-center gap-2">
 Comptes Administrateurs
</h3>
<form id="formCreateAdmin" class="grid grid-cols-1 sm:grid-cols-3 gap-3 bg-slate-50 p-4 rounded-xl border border-slate-200">

<label class="block text-xs font-bold text-slate-600 mb-1">Identifiant</label>
<input type="text" id="newAdminName" required placeholder="Ex: Admin_QA" class="w-full px-3 py-2 border rounded-xl text-xs bg-white">

<label class="block text-xs font-bold text-slate-600 mb-1">Mot de passe</label>
<input type="password" id="newAdminPassword" required placeholder="••••••••" class="w-full px-3 py-2 border rounded-xl text-xs bg-white">

<button type="submit" class="w-full py-2 bg-slate-900 hover:bg-black text-amber-300 font-bold rounded-xl text-xs transition">
+ Ajouter cet Admin
</button>

</form>

<table class="w-full text-left text-sm">
<thead class="bg-slate-50 text-slate-500 uppercase text-xs">
<tr>
<th class="px-4 py-2">Administrateur</th>

```
              <th class="px-4 py-2">Date création</th>
              <th class="px-4 py-2 text-right">Action</th>
            </tr>
          </thead>
          <tbody id="adminUsersTable" class="divide-y divide-slate-100"></tbody>
        </table>
      </div>

      <!-- Paramétrage Métier -->
      <div class="grid grid-cols-1 md:grid-cols-3 gap-6">
        <div class="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm space-y-4">
          <h3 class="font-bold text-slate-800 text-sm flex items-center gap-2">
            <i class="fa-solid fa-briefcase text-indigo-600"></i> Activités
          </h3>
          <div class="flex gap-2">
            <input type="text" id="newActivityInput" placeholder="Ex: SAV N2..." class="flex-1 px-3
py-1.5 border rounded-xl text-xs">
            <button onclick="addActivity()" class="px-3 py-1.5 bg-indigo-600 text-white font-bold
rounded-xl text-xs">Ajouter</button>
          </div>
          <ul id="adminActivityList" class="divide-y divide-slate-100 max-h-60
overflow-y-auto"></ul>
        </div>

        <div class="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm space-y-4">
          <h3 class="font-bold text-slate-800 text-sm flex items-center gap-2">
            <i class="fa-solid fa-bullseye text-indigo-600"></i> Cibles de Clarification
          </h3>
          <div class="flex gap-2">
            <input type="text" id="newClarifInput" placeholder="Ex: Tampon..." class="flex-1 px-3
py-1.5 border rounded-xl text-xs">
            <button onclick="addClarifType()" class="px-3 py-1.5 bg-indigo-600 text-white font-bold
rounded-xl text-xs">Ajouter</button>
          </div>
          <ul id="adminClarifList" class="divide-y divide-slate-100 max-h-60
overflow-y-auto"></ul>
        </div>

        <div class="bg-white p-5 rounded-2xl border border-slate-200 shadow-sm space-y-4">
          <h3 class="font-bold text-slate-800 text-sm flex items-center gap-2">
            <i class="fa-solid fa-folder-tree text-indigo-600"></i> Types de Documents
          </h3>
          <div class="flex gap-2">
```

```
            <input type="text" id="newDocTypeInput" placeholder="Nouveau Type..." class="flex-1
px-3 py-1.5 border rounded-xl text-xs">
            <button onclick="addDocType()" class="px-3 py-1.5 bg-indigo-600 text-white font-bold
rounded-xl text-xs">Ajouter</button>
          </div>
          <div id="adminDocTypeList" class="space-y-3 max-h-60 overflow-y-auto"></div>
        </div>
      </div>
    </section>
  </div>
 </main>

 <!-- ================= MODALE DEMANDE AGENT ================= -->
 <div id="modalRequest" class="fixed inset-0 bg-slate-900 bg-opacity-60 z-50 flex items-center
justify-center p-4 hidden">
  <div class="bg-white w-full max-w-lg rounded-3xl shadow-2xl border border-slate-100
overflow-hidden">
    <div class="bg-indigo-950 px-6 py-4 flex items-center justify-between text-white">
      <h3 class="font-bold text-base flex items-center gap-2">
        <i class="fa-solid fa-hand-holding-hand text-rose-400"></i> Nouvelle Demande
d'Assistance
      </h3>
      <button onclick="closeRequestModal()" class="text-indigo-300 hover:text-white
text-xl">&times;</button>
    </div>

    <form id="formNewRequest" class="p-6 space-y-4">
      <div>
        <label class="block text-xs font-bold uppercase tracking-wider text-slate-600 mb-1">
          Lien Direct du Document <span class="text-rose-500">*</span>
        </label>
        <input type="url" id="reqDocLink" required
placeholder="https://ged.entreprise.com/view/99281" class="w-full px-3 py-2 border rounded-xl
text-sm focus:ring-2 focus:ring-indigo-500">
      </div>

      <div class="grid grid-cols-1 sm:grid-cols-2 gap-4">
        <div>
          <label class="block text-xs font-bold uppercase tracking-wider text-slate-600
mb-1">Type de Document</label>
          <select id="reqDocType" required onchange="handleDocTypeChange()" class="w-full
px-3 py-2 border rounded-xl text-sm bg-white"></select>
        </div>
        <div>
```

```
          <label class="block text-xs font-bold uppercase tracking-wider text-slate-600
mb-1">Sous-type</label>
          <select id="reqSubType" required class="w-full px-3 py-2 border rounded-xl text-sm
bg-white"></select>
        </div>
      </div>

      <div class="p-4 bg-indigo-50/50 rounded-2xl border border-indigo-100 space-y-3">
        <div>
          <label class="block text-xs font-bold uppercase tracking-wider text-indigo-950 mb-1">
            <i class="fa-solid fa-bullseye text-rose-500 mr-1"></i> Ma question concerne
particulièrement :
          </label>
          <select id="reqClarifTarget" required onchange="handleClarifTargetChange()"
class="w-full px-3 py-2 border border-indigo-200 rounded-xl text-sm font-bold bg-white
text-indigo-900"></select>
        </div>

        <div>
          <label id="lblClarifDetail" class="block text-xs font-semibold text-slate-600 mb-1">
            Valeur ou observation sur la pièce :
          </label>
          <input type="text" id="reqClarifValue" placeholder="Ex: Original vs Copie / 12-05-2024 /
N° 88192" class="w-full px-3 py-2 border rounded-xl text-xs bg-white">
        </div>

        <div>
          <label class="block text-xs font-semibold text-slate-600 mb-1">Question précise pour le
Support</label>
          <textarea id="reqDocNotes" rows="2" placeholder="Précisez votre interrogation pour
orienter le support..." class="w-full px-3 py-2 border rounded-xl text-xs bg-white"></textarea>
        </div>
      </div>

      <div class="flex items-center justify-end space-x-3 pt-2">
        <button type="button" onclick="closeRequestModal()" class="px-4 py-2 border
border-slate-300 rounded-xl text-sm font-semibold hover:bg-slate-50">Annuler</button>
        <button type="submit" class="px-5 py-2 bg-rose-600 hover:bg-rose-700 text-white
font-bold rounded-xl text-sm shadow">
          Transmettre au Support
        </button>
      </div>
    </form>
  </div>
```

```
  </div>

  <div id="toastContainer" class="fixed bottom-4 right-4 z-50 space-y-2
pointer-events-none"></div>

  <!-- ================ JAVASCRIPT ================ -->
  <script>
    /* ÉTAT INITIAL */
    const DEFAULT_DATA = {
      admins: [],
      activities: ['KYC Particuliers', 'Souscription Entreprise', 'Traitement Sinistres', 'Facturation &
Recouvrement'],
      clarifTargets: ['Nature du document', 'Date du document', 'Numéro du document'],
      docCategories: [
        { id: 'identite', name: "Pièce d'Identité", subTypes: ["Passeport", "CNI (Carte Nationale)",
"Titre de Séjour"] },
        { id: 'domicile', name: "Justificatif de Domicile", subTypes: ["Facture Énergie (-3 mois)",
"Avis d'Imposition"] }
      ],
      tickets: [],
      currentUser: null,
      firebaseConfig: ""
    };

    function loadState() {
      const data = localStorage.getItem('helpdesk_web_state');
      return data ? JSON.parse(data) : DEFAULT_DATA;
    }

    let appState = loadState();
    let firebaseDb = null;

    /* CANAL INTER-ONGLETS (Local Broadcast) */
    const liveChannel = new BroadcastChannel('helpdesk_web_sync');
    liveChannel.onmessage = (event) => {
      if (event.data && event.data.type === 'REFRESH') {
        appState = loadState();
        refreshCurrentViews();
        if (event.data.newTicket && appState.currentUser && (appState.currentUser.role ===
'support' || appState.currentUser.role === 'admin')) {
          playChime();
          showToast("Nouvelle alerte reçue !", `Demande #${event.data.newTicket.id} par
${event.data.newTicket.agentName}`);
        }
```

```
  }
};

/* INITIALISATION FIREBASE CLOUD (Pour synchronisation collaborative en direct) */
function initFirebase() {
  if (appState.firebaseConfig) {
    try {
      if (!firebase.apps.length) {
        let configObj;
        if (appState.firebaseConfig.startsWith('http')) {
          configObj = { databaseURL: appState.firebaseConfig };
        } else {
          configObj = JSON.parse(appState.firebaseConfig);
        }
        firebase.initializeApp(configObj);
      }
      firebaseDb = firebase.database();

      // Écoute temps réel des tickets
      firebaseDb.ref('tickets').on('value', (snapshot) => {
        const data = snapshot.val();
        if (data) {
          const remoteTickets = Object.values(data);
          // Détecter un nouveau ticket
          if (remoteTickets.length > appState.tickets.length && appState.currentUser &&
(appState.currentUser.role === 'support' || appState.currentUser.role === 'admin')) {
            playChime();
            showToast("Nouvelle demande reçue !", "Mise à jour en temps réel.");
          }
          appState.tickets = remoteTickets.sort((a,b) => new Date(b.createdAt) - new
Date(a.createdAt));
          localStorage.setItem('helpdesk_web_state', JSON.stringify(appState));
          refreshCurrentViews();
        }
      });

      // Écoute temps réel des paramètres
      firebaseDb.ref('settings').on('value', (snapshot) => {
        const settings = snapshot.val();
        if (settings) {
          if (settings.activities) appState.activities = settings.activities;
          if (settings.clarifTargets) appState.clarifTargets = settings.clarifTargets;
          if (settings.docCategories) appState.docCategories = settings.docCategories;
          if (settings.admins) appState.admins = settings.admins;
```

```
            localStorage.setItem('helpdesk_web_state', JSON.stringify(appState));
            refreshCurrentViews();
          }
        });

        document.getElementById('syncDot').className = "w-2 h-2 rounded-full bg-emerald-500
animate-pulse";
        document.getElementById('syncText').innerText = "Cloud En Ligne";
        if (document.getElementById('adminSyncBadge')) {
          document.getElementById('adminSyncBadge').innerText = "Connecté";
          document.getElementById('adminSyncBadge').className = "text-xs font-bold px-2.5
py-1 rounded-full bg-emerald-100 text-emerald-800";
        }
      } catch (e) {
        console.warn("Erreur connexion Firebase :", e);
      }
    }
  }

  function saveState(state, notify = false, newTicket = null) {
    localStorage.setItem('helpdesk_web_state', JSON.stringify(state));
    if (firebaseDb) {
      if (newTicket) {
        firebaseDb.ref('tickets/' + newTicket.id).set(newTicket);
      }
      firebaseDb.ref('settings').set({
        activities: state.activities,
        clarifTargets: state.clarifTargets,
        docCategories: state.docCategories,
        admins: state.admins
      });
    }
    if (notify) {
      liveChannel.postMessage({ type: 'REFRESH', newTicket });
    }
  }

  function saveFirebaseConfig() {
    const val = document.getElementById('firebaseConfigInput').value.trim();
    appState.firebaseConfig = val;
    saveState(appState);
    showToast("Configuration enregistrée", "Initialisation de la synchronisation...");
    setTimeout(() => location.reload(), 1000);
  }
```

```
    /* HACHAGE MOT DE PASSE (SHA-256) */
    async function hashPassword(plainText) {
      const encoder = new TextEncoder();
      const data = encoder.encode(plainText);
      const hashBuffer = await crypto.subtle.digest('SHA-256', data);
      return Array.from(new Uint8Array(hashBuffer)).map(b => b.toString(16).padStart(2,
'0')).join('');
    }

    /* Bip Sonore Web Audio */
    function playChime() {
      try {
        const ctx = new (window.AudioContext || window.webkitAudioContext)();
        const now = ctx.currentTime;
        const osc = ctx.createOscillator();
        const gain = ctx.createGain();
        osc.frequency.setValueAtTime(880, now);
        gain.gain.setValueAtTime(0.25, now);
        gain.gain.exponentialRampToValueAtTime(0.001, now + 0.3);
        osc.connect(gain);
        gain.connect(ctx.destination);
        osc.start(now);
        osc.stop(now + 0.3);

        const osc2 = ctx.createOscillator();
        const gain2 = ctx.createGain();
        osc2.frequency.setValueAtTime(1174.66, now + 0.12);
        gain2.gain.setValueAtTime(0.3, now + 0.12);
        gain2.gain.exponentialRampToValueAtTime(0.001, now + 0.5);
        osc2.connect(gain2);
        gain2.connect(ctx.destination);
        osc2.start(now + 0.12);
        osc2.stop(now + 0.5);
      } catch (e) {}
    }

    function testSoundNotification() {
      playChime();
      showToast("Test Sonore", "Le signal d'alerte fonctionne.");
    }

    function showToast(title, msg) {
      const toast = document.createElement('div');
```

```
    toast.className = 'bg-slate-900 text-white px-4 py-3 rounded-2xl shadow-xl flex
items-center gap-3 border-l-4 border-rose-500 pointer-events-auto transform transition
duration-300';
    toast.innerHTML = `
      <div class="text-rose-400 text-lg"><i class="fa-solid fa-bell"></i></div>
      <div>
        <p class="font-bold text-xs uppercase tracking-wider">${title}</p>
        <p class="text-xs text-slate-300">${msg}</p>
      </div>
    `;
    document.getElementById('toastContainer').appendChild(toast);
    setTimeout(() => toast.remove(), 4000);
  }

  function copyCurrentUrl() {
    navigator.clipboard.writeText(window.location.href);
    showToast("Lien copié !", "Vous pouvez l'envoyer à vos collègues.");
  }

  /* GESTION DE L'AFFICHAGE */
  function showView(viewName) {
    ['viewLoginSelector', 'viewOperativeAuth', 'viewAdminAuth', 'viewWorkspace'].forEach(id =>
{
      document.getElementById(id).classList.add('hidden');
    });
    if (viewName === 'loginSelector')
document.getElementById('viewLoginSelector').classList.remove('hidden');
    if (viewName === 'operativeAuth')
document.getElementById('viewOperativeAuth').classList.remove('hidden');
    if (viewName === 'adminAuth')
document.getElementById('viewAdminAuth').classList.remove('hidden');
    if (viewName === 'workspace')
document.getElementById('viewWorkspace').classList.remove('hidden');
  }

  function showOperativeLogin() {
    renderActivitySelects();
    showView('operativeAuth');
  }

  function showAdminLogin() {
    const notice = document.getElementById('adminFirstSetupNotice');
    const submitBtn = document.getElementById('btnAdminSubmit');
    if (appState.admins.length === 0) {
```

```
      notice.classList.remove('hidden');
      submitBtn.innerText = "Créer le premier Admin & Se connecter";
    } else {
      notice.classList.add('hidden');
      submitBtn.innerText = "Se connecter";
    }
    showView('adminAuth');
  }

  /* CONNEXION */
  document.getElementById('formOperativeAuth').addEventListener('submit', (e) => {
    e.preventDefault();
    const name = document.getElementById('opName').value.trim();
    const activity = document.getElementById('opActivity').value;
    const role = document.querySelector('input[name="opRole"]:checked').value;
    appState.currentUser = { name, activity, role };
    saveState(appState);
    launchWorkspace();
  });

  document.getElementById('formAdminAuth').addEventListener('submit', async (e) => {
    e.preventDefault();
    const name = document.getElementById('adminAuthName').value.trim();
    const pass = document.getElementById('adminAuthPassword').value;
    const passHash = await hashPassword(pass);

    if (appState.admins.length === 0) {
      const firstAdmin = { id: 'ADM-' + Date.now(), name, passHash, createdAt: new
Date().toISOString() };
      appState.admins.push(firstAdmin);
      appState.currentUser = { name: firstAdmin.name, role: 'admin', activity: 'Administration' };
      saveState(appState);
      launchWorkspace();
      showToast("Compte Admin Créé", `Bienvenue ${name}`);
      return;
    }

    const found = appState.admins.find(a => a.name.toLowerCase() === name.toLowerCase()
&& a.passHash === passHash);
    if (found) {
      appState.currentUser = { name: found.name, role: 'admin', activity: 'Administration' };
      saveState(appState);
      launchWorkspace();
    } else {
```

```
      alert("Identifiant ou mot de passe incorrect.");
    }
  });

  document.getElementById('formCreateAdmin').addEventListener('submit', async (e) => {
    e.preventDefault();
    const name = document.getElementById('newAdminName').value.trim();
    const pass = document.getElementById('newAdminPassword').value;
    if (appState.admins.some(a => a.name.toLowerCase() === name.toLowerCase())) {
      alert("Un administrateur porte déjà ce nom.");
      return;
    }
    const passHash = await hashPassword(pass);
    appState.admins.push({ id: 'ADM-' + Date.now(), name, passHash, createdAt: new
Date().toISOString() });
    saveState(appState);
    renderAdminUsersTable();
    document.getElementById('formCreateAdmin').reset();
    showToast("Nouvel Admin", `${name} a été ajouté.`);
  });

  function removeAdmin(id) {
    if (appState.admins.length <= 1) {
      alert("Impossible de supprimer le dernier compte administrateur.");
      return;
    }
    if (confirm("Supprimer cet accès administrateur ?")) {
      appState.admins = appState.admins.filter(a => a.id !== id);
      saveState(appState);
      renderAdminUsersTable();
    }
  }

  function renderAdminUsersTable() {
    const tbody = document.getElementById('adminUsersTable');
    tbody.innerHTML = appState.admins.map(a => `
      <tr class="hover:bg-slate-50">
        <td class="px-4 py-2.5 font-bold text-slate-800 text-xs">${a.name}</td>
        <td class="px-4 py-2.5 text-xs text-slate-500">${new
Date(a.createdAt).toLocaleDateString()}</td>
        <td class="px-4 py-2.5 text-right">
          ${appState.currentUser.name !== a.name ? `
            <button onclick="removeAdmin('${a.id}')" class="text-rose-600 hover:text-rose-800
font-bold text-xs">Supprimer</button>
```

```
        ` : '<span class="text-xs text-slate-400 italic">Votre session</span>'}
      </td>
    </tr>
  `).join('');
}

/* ESPACE DE TRAVAIL CONNECTÉ */
function launchWorkspace() {
  showView('workspace');
  renderHeaderNav();

  const role = appState.currentUser.role;
  const tabAgent = document.getElementById('tabBtnAgent');
  const tabSupport = document.getElementById('tabBtnSupport');
  const tabStats = document.getElementById('tabBtnStats');
  const tabAdmin = document.getElementById('tabBtnAdmin');

  if (role === 'agent') {
    tabAgent.classList.remove('hidden');
    tabSupport.classList.add('hidden');
    tabStats.classList.add('hidden');
    tabAdmin.classList.add('hidden');
    switchTab('agent');
  } else if (role === 'support') {
    tabAgent.classList.remove('hidden');
    tabSupport.classList.remove('hidden');
    tabStats.classList.remove('hidden');
    tabAdmin.classList.add('hidden');
    switchTab('support');
  } else if (role === 'admin') {
    tabAgent.classList.remove('hidden');
    tabSupport.classList.remove('hidden');
    tabStats.classList.remove('hidden');
    tabAdmin.classList.remove('hidden');
    switchTab('support');
  }
  updatePendingBadge();
}

function logout() {
  appState.currentUser = null;
  saveState(appState);
  showView('loginSelector');
  renderHeaderNav();
```

```
  }

  function renderHeaderNav() {
    const container = document.getElementById('headerUserNav');
    const roleIndicator = document.getElementById('roleIndicator');

    if (!appState.currentUser) {
      roleIndicator.innerText = "Portail d'Assistance";
      container.innerHTML = `
        <button onclick="showOperativeLogin()" class="px-3 py-1.5 bg-indigo-700
hover:bg-indigo-600 text-white text-xs font-semibold rounded-xl">
          Agent / QA
        </button>
        <button onclick="showAdminLogin()" class="px-3 py-1.5 bg-amber-500
hover:bg-amber-400 text-slate-900 text-xs font-bold rounded-xl">
          Admin
        </button>
      `;
      return;
    }

    const labels = {
      agent: '<span class="bg-indigo-800 text-indigo-200 px-2 py-0.5 rounded
text-xs">Agent</span>',
      support: '<span class="bg-emerald-800 text-emerald-100 px-2 py-0.5 rounded text-xs">QA
/ Trainer</span>',
      admin: '<span class="bg-amber-400 text-slate-950 font-bold px-2 py-0.5 rounded
text-xs">Admin</span>'
    };

    roleIndicator.innerHTML = `Connecté : <strong>
appState.currentUser.name < /strong > ({appState.currentUser.activity})`;
    container.innerHTML = `
      <div class="text-right hidden sm:block">
        <p class="text-xs font-bold">${appState.currentUser.name}
${labels[appState.currentUser.role]}</p>
      </div>
      <button onclick="logout()" title="Déconnexion" class="bg-indigo-800 hover:bg-rose-600
text-white p-2 rounded-xl text-xs transition">
        <i class="fa-solid fa-power-off"></i>
      </button>
    `;
  }
```

```
  function switchTab(tab) {
    if ((tab === 'stats' || tab === 'support') && appState.currentUser.role === 'agent') tab =
'agent';
    if (tab === 'admin' && appState.currentUser.role !== 'admin') tab = 'agent';

    ['agent', 'support', 'stats', 'admin'].forEach(t => {
      const sec = document.getElementById(`tab${capitalize(t)}`);
      if (sec) sec.classList.add('hidden');
      const btn = document.getElementById(`tabBtn${capitalize(t)}`);
      if (btn) {
        btn.classList.remove('bg-indigo-600', 'text-white');
        btn.classList.add('text-slate-600', 'hover:bg-slate-200');
      }
    });

    document.getElementById(`tab${capitalize(tab)}`).classList.remove('hidden');
    const activeBtn = document.getElementById(`tabBtn${capitalize(tab)}`);
    activeBtn.classList.remove('text-slate-600', 'hover:bg-slate-200');
    activeBtn.classList.add('bg-indigo-600', 'text-white');

    refreshCurrentViews();
  }

  function capitalize(s) { return s.charAt(0).toUpperCase() + s.slice(1); }

  function refreshCurrentViews() {
    renderAgentTickets();
    if (appState.currentUser && appState.currentUser.role !== 'agent') {
      renderSupportTickets();
      renderStats();
    }
    if (appState.currentUser && appState.currentUser.role === 'admin') {
      renderAdminView();
    }
    updatePendingBadge();
  }

  function renderActivitySelects() {
    const select = document.getElementById('opActivity');
    const filter = document.getElementById('filterActivity');
    select.innerHTML = appState.activities.map(a => `<option value="a" >{a}</option>`).join('');
    if (filter) {
      filter.innerHTML = `<option value="ALL">Toutes les activités</option>` +
appState.activities.map(a => `<option value="a" >{a}</option>`).join('');
```

```
    }
  }

  function renderDocTypeSelects() {
    const docTypeSelect = document.getElementById('reqDocType');
    docTypeSelect.innerHTML = appState.docCategories.map(cat => `<option value="
cat.name" >{cat.name}</option>`).join('');
    handleDocTypeChange();

    const clarifSelect = document.getElementById('reqClarifTarget');
    clarifSelect.innerHTML = appState.clarifTargets.map(t => `<option value="t" >
{t}</option>`).join('');
    handleClarifTargetChange();
  }

  function handleDocTypeChange() {
    const selectedTypeName = document.getElementById('reqDocType').value;
    const subTypeSelect = document.getElementById('reqSubType');
    const category = appState.docCategories.find(c => c.name === selectedTypeName);
    if (category && category.subTypes) {
      subTypeSelect.innerHTML = category.subTypes.map(st => `<option value="st" >
{st}</option>`).join('');
    } else {
      subTypeSelect.innerHTML = `<option value="Standard">Standard</option>`;
    }
  }

  function handleClarifTargetChange() {
    const target = document.getElementById('reqClarifTarget').value;
    const lbl = document.getElementById('lblClarifDetail');
    const input = document.getElementById('reqClarifValue');
    lbl.innerText = `Valeur concernant [${target}] :`;
    if (target.toLowerCase().includes('date')) input.placeholder = "Ex: 12/04/2025 ou date
introuvable";
    else if (target.toLowerCase().includes('numéro')) input.placeholder = "Ex: FAC-9901 ou
numéro illisible";
    else input.placeholder = "Ex: Tampon manquant, copie non certifiée...";
  }

  /* DEMANDE D'AIDE AGENT */
  function openRequestModal() {
    renderDocTypeSelects();
    document.getElementById('modalRequest').classList.remove('hidden');
  }
```

```
  function closeRequestModal() {
    document.getElementById('modalRequest').classList.add('hidden');
    document.getElementById('formNewRequest').reset();
  }

  document.getElementById('formNewRequest').addEventListener('submit', (e) => {
    e.preventDefault();
    const newTicket = {
      id: 'TICK-' + Date.now().toString().slice(-5),
      agentName: appState.currentUser.name,
      activity: appState.currentUser.activity,
      docLink: document.getElementById('reqDocLink').value,
      docType: document.getElementById('reqDocType').value,
      subType: document.getElementById('reqSubType').value,
      clarifTarget: document.getElementById('reqClarifTarget').value,
      clarifValue: document.getElementById('reqClarifValue').value || 'Non précisé',
      notes: document.getElementById('reqDocNotes').value || '',
      status: 'PENDING',
      assignedTo: null,
      createdAt: new Date().toISOString()
    };

    appState.tickets.unshift(newTicket);
    saveState(appState, true, newTicket);

    closeRequestModal();
    renderAgentTickets();
    showToast("Demande envoyée !", `Ticket #${newTicket.id} transmis au Support QA.`);
  });

  /* RENDU AGENT (SES TICKETS UNIQUEMENT) */
  function renderAgentTickets() {
    const tbody = document.getElementById('agentTicketsTable');
    if (!tbody) return;
    const myTickets = appState.tickets.filter(t => t.agentName === appState.currentUser.name);

    if (myTickets.length === 0) {
      tbody.innerHTML = `<tr><td colspan="7" class="px-6 py-8 text-center
text-slate-400">Aucune demande enregistrée.</td></tr>`;
      return;
    }

    tbody.innerHTML = myTickets.map(t => {
```

```
      let badge = `<span class="bg-amber-100 text-amber-800 text-xs px-2.5 py-1 rounded-full font-bold">En attente</span>`;
      if (t.status === 'PROGRESS') badge = `<span class="bg-blue-100 text-blue-800 text-xs px-2.5 py-1 rounded-full font-bold">En cours</span>`;
      if (t.status === 'RESOLVED') badge = `<span class="bg-emerald-100 text-emerald-800 text-xs px-2.5 py-1 rounded-full font-bold">Résolu</span>`;

      return `
        <tr class="hover:bg-slate-50">
          <td class="px-6 py-4 text-xs text-slate-500 whitespace-nowrap">${new Date(t.createdAt).toLocaleTimeString([], {hour:'2-digit', minute:'2-digit'})}</td>
          <td class="px-6 py-4">
            <a href="${t.docLink}" target="_blank" class="text-indigo-600 hover:underline font-bold flex items-center gap-1.5 text-xs">
              <i class="fa-solid fa-arrow-up-right-from-square"></i> Consulter
            </a>
          </td>
          <td class="px-6 py-4">
            <p class="font-bold text-slate-800">${t.docType}</p>
            <p class="text-xs text-slate-500">${t.subType}</p>
          </td>
          <td class="px-6 py-4">
            <span class="bg-indigo-50 text-indigo-700 font-bold px-2 py-0.5 rounded text-xs">${t.clarifTarget}</span>
            <p class="text-xs text-slate-600 mt-0.5">${t.clarifValue}</p>
          </td>
          <td class="px-6 py-4">${badge}</td>
          <td class="px-6 py-4 text-xs font-semibold text-slate-700">
            ${t.assignedTo ? `<i class="fa-solid fa-user-check text-indigo-500 mr-1"></i> ${t.assignedTo}` : '<span class="text-slate-400 italic">En attente</span>'}
          </td>
          <td class="px-6 py-4">
            ${t.status !== 'RESOLVED' ? `
              <button onclick="resolveTicket('${t.id}')" class="text-xs bg-emerald-50 text-emerald-700 border border-emerald-300 hover:bg-emerald-100 px-3 py-1 rounded-lg font-bold">
                Clôturer
              </button>
            ` : '<i class="fa-solid fa-check-double text-emerald-500"></i>'}
          </td>
        </tr>
      `;
    }).join('');
  }
```

```
   /* RENDU FILE GLOBALE SUPPORT */
   function renderSupportTickets() {
     const container = document.getElementById('supportTicketsContainer');
     if (!container) return;
     const filter = document.getElementById('filterActivity') ?
document.getElementById('filterActivity').value : 'ALL';

     let tickets = appState.tickets;
     if (filter !== 'ALL') tickets = tickets.filter(t => t.activity === filter);

     if (tickets.length === 0) {
       container.innerHTML = `<div class="col-span-full text-center py-12 bg-white rounded-2xl
border border-dashed border-slate-300 text-slate-400">Aucun ticket dans cette
catégorie.</div>`;
       return;
     }

     container.innerHTML = tickets.map(t => {
       let borderClass = 'border-amber-300 bg-amber-50/20';
       if (t.status === 'PROGRESS') borderClass = 'border-blue-400 bg-blue-50/20';
       if (t.status === 'RESOLVED') borderClass = 'border-slate-200 bg-slate-50 opacity-70';

       return `
         <div class="bg-white rounded-2xl border-2 ${borderClass} p-5 shadow-sm flex flex-col
justify-between space-y-4">
           <div>
             <div class="flex items-center justify-between mb-2">
               <span class="text-xs font-bold px-2 py-0.5 rounded bg-slate-100
text-slate-700">${t.activity}</span>
               <span class="text-xs text-slate-400">${new Date(t.createdAt).toLocaleTimeString([],
{hour:'2-digit', minute:'2-digit'})}</span>
             </div>
             <h4 class="font-black text-slate-900 flex items-center justify-between">
               <span>${t.docType}</span>
               <span class="text-xs text-indigo-600 font-normal">#${t.id}</span>
             </h4>
             <p class="text-xs font-bold text-slate-500 mt-0.5">${t.subType}</p>

             <div class="mt-3 p-3 bg-indigo-50/60 rounded-xl text-xs space-y-1.5 border
border-indigo-100">
               <div class="flex items-center gap-1.5">
                 <span class="text-xs uppercase font-black text-indigo-900">Objet :</span>
```

```
              <span class="bg-indigo-600 text-white font-bold px-2 py-0.5 rounded
text-xs">${t.clarifTarget}</span>
            </div>
            <p><strong class="text-slate-700">Précision :</strong> ${t.clarifValue}</p>
            <p><strong class="text-slate-700">Agent :</strong> ${t.agentName}</p>
            ${t.notes ? `<p class="mt-2 text-slate-600 italic border-t border-indigo-200/50
pt-1">"${t.notes}"</p>` : ''}
          </div>

          <div class="mt-3">
            <a href="${t.docLink}" target="_blank" class="w-full py-2 bg-indigo-600
hover:bg-indigo-700 text-white font-bold text-xs rounded-xl flex items-center justify-center
gap-1.5 transition">
              <i class="fa-solid fa-file-lines"></i> Ouvrir le document à vérifier
            </a>
          </div>
        </div>

        <div class="pt-3 border-t border-slate-100 flex items-center justify-between">
          <div>
            ${t.assignedTo ? `<p class="text-xs text-slate-500">Pris par: <strong
class="text-slate-800">${t.assignedTo}</strong></p>` : `<span class="text-xs text-amber-600
font-bold"><i class="fa-solid fa-hourglass-start"></i> En attente</span>`}
          </div>
          <div class="space-x-1">
            ${t.status === 'PENDING' ? `
              <button onclick="assignTicketToMe('${t.id}')" class="px-3 py-1.5 bg-blue-600
hover:bg-blue-700 text-white font-bold rounded-xl text-xs">
                Prendre en charge
              </button>
            ` : ''}
            ${t.status === 'PROGRESS' ? `
              <button onclick="resolveTicket('${t.id}')" class="px-3 py-1.5 bg-emerald-600
hover:bg-emerald-700 text-white font-bold rounded-xl text-xs">
                Résoudre
              </button>
            ` : ''}
            ${t.status === 'RESOLVED' ? `<span class="text-xs text-emerald-600 font-bold"><i
class="fa-solid fa-check"></i> Clôturé</span>` : ''}
          </div>
        </div>
      </div>
    `;
  }).join('');
```

```
  }

  function assignTicketToMe(id) {
    const ticket = appState.tickets.find(t => t.id === id);
    if (ticket) {
      ticket.status = 'PROGRESS';
      ticket.assignedTo = appState.currentUser.name;
      if (firebaseDb) firebaseDb.ref('tickets/' + id).update({ status: 'PROGRESS', assignedTo:
appState.currentUser.name });
      saveState(appState, true);
      renderSupportTickets();
    }
  }

  function resolveTicket(id) {
    const ticket = appState.tickets.find(t => t.id === id);
    if (ticket) {
      ticket.status = 'RESOLVED';
      if (!ticket.assignedTo) ticket.assignedTo = appState.currentUser.name;
      if (firebaseDb) firebaseDb.ref('tickets/' + id).update({ status: 'RESOLVED', assignedTo:
ticket.assignedTo });
      saveState(appState, true);
      refreshCurrentViews();
    }
  }

  function updatePendingBadge() {
    const count = appState.tickets.filter(t => t.status === 'PENDING').length;
    const b = document.getElementById('badgePendingCount');
    if (b) b.innerText = count;
  }

  /* STATISTIQUES & DONNÉES */
  function renderStats() {
    const tickets = appState.tickets;
    document.getElementById('statTotalRequests').innerText = tickets.length;
    document.getElementById('statPendingRequests').innerText = tickets.filter(t => t.status ===
'PENDING').length;
    document.getElementById('statProgressRequests').innerText = tickets.filter(t => t.status ===
'PROGRESS').length;
    document.getElementById('statResolvedRequests').innerText = tickets.filter(t => t.status ===
'RESOLVED').length;

    const qaMap = {};
```

```
    tickets.forEach(t => {
      if (t.assignedTo) {
        if (!qaMap[t.assignedTo]) qaMap[t.assignedTo] = { progress: 0, resolved: 0 };
        if (t.status === 'PROGRESS') qaMap[t.assignedTo].progress++;
        if (t.status === 'RESOLVED') qaMap[t.assignedTo].resolved++;
      }
    });

    document.getElementById('statsQaTable').innerHTML = Object.keys(qaMap).map(qa => `
      <tr>
        <td class="px-4 py-3 font-bold text-slate-800">${qa}</td>
        <td class="px-4 py-3 text-blue-600 font-bold">${qaMap[qa].progress}</td>
        <td class="px-4 py-3 text-emerald-600 font-bold">${qaMap[qa].resolved}</td>
        <td class="px-4 py-3 text-right font-black">${qaMap[qa].progress +
qaMap[qa].resolved}</td>
      </tr>
    `).join('') || `<tr><td colspan="4" class="px-4 py-4 text-center text-slate-400">Aucun
traitement support.</td></tr>`;

    const agentMap = {};
    tickets.forEach(t => {
      if (!agentMap[t.agentName]) agentMap[t.agentName] = { count: 0, activity: t.activity };
      agentMap[t.agentName].count++;
    });

    document.getElementById('statsAgentTable').innerHTML =
Object.keys(agentMap).sort((a,b) => agentMap[b].count - agentMap[a].count).map(agent => `
      <tr>
        <td class="px-4 py-3 font-bold text-slate-800">${agent}</td>
        <td class="px-4 py-3 text-xs text-slate-500">${agentMap[agent].activity}</td>
        <td class="px-4 py-3 text-right font-black text-indigo-600">${agentMap[agent].count}</td>
      </tr>
    `).join('') || `<tr><td colspan="3" class="px-4 py-4 text-center text-slate-400">Aucune
donnée.</td></tr>`;
  }

  /* CONFIGURATION ADMIN */
  function renderAdminView() {
    renderAdminUsersTable();
    if (appState.firebaseConfig) {
      document.getElementById('firebaseConfigInput').value = appState.firebaseConfig;
    }
```

```
    document.getElementById('adminActivityList').innerHTML = appState.activities.map((act,
idx) => `
      <li class="py-2 flex items-center justify-between text-xs">
        <span class="font-medium text-slate-700">${act}</span>
        <button onclick="removeActivity(${idx})" class="text-rose-500 hover:text-rose-700
font-bold">Supprimer</button>
      </li>
    `).join('');

    document.getElementById('adminClarifList').innerHTML = appState.clarifTargets.map((item,
idx) => `
      <li class="py-2 flex items-center justify-between text-xs">
        <span class="font-bold text-indigo-900">${item}</span>
        <button onclick="removeClarifType(${idx})" class="text-rose-500 hover:text-rose-700
font-bold">Supprimer</button>
      </li>
    `).join('');

    document.getElementById('adminDocTypeList').innerHTML =
appState.docCategories.map((cat, catIdx) => `
      <div class="border rounded-xl p-3 bg-slate-50 space-y-2">
        <div class="flex items-center justify-between">
          <h4 class="font-bold text-slate-800 text-xs">${cat.name}</h4>
          <button onclick="removeDocCategory(${catIdx})" class="text-rose-500 text-xs
font-bold">Supprimer</button>
        </div>
        <div class="flex flex-wrap gap-1">
          ${cat.subTypes.map((st, subIdx) => `
            <span class="bg-white border text-[11px] px-2 py-0.5 rounded-full text-slate-700 flex
items-center gap-1 font-medium">
              st < buttononclick = "removeSubType({catIdx}, ${subIdx})" class="text-slate-400
hover:text-rose-500">&times;</button>
            </span>
          `).join('')}
        </div>
        <div class="flex gap-1 pt-1">
          <input type="text" id="subInput_${catIdx}" placeholder="Sous-catégorie..." class="text-xs
px-2 py-1 border rounded-lg flex-1 bg-white">
          <button onclick="addSubType(${catIdx})" class="text-xs px-2.5 py-1 bg-slate-800
text-white rounded-lg font-bold">OK</button>
        </div>
      </div>
    `).join('');
  }
```

```
function addActivity() {
  const inp = document.getElementById('newActivityInput');
  const val = inp.value.trim();
  if (val && !appState.activities.includes(val)) {
    appState.activities.push(val);
    saveState(appState, true);
    renderActivitySelects();
    renderAdminView();
    inp.value = '';
  }
}
function removeActivity(idx) {
  appState.activities.splice(idx, 1);
  saveState(appState, true);
  renderActivitySelects();
  renderAdminView();
}
function addClarifType() {
  const inp = document.getElementById('newClarifInput');
  const val = inp.value.trim();
  if (val && !appState.clarifTargets.includes(val)) {
    appState.clarifTargets.push(val);
    saveState(appState, true);
    renderAdminView();
    inp.value = '';
  }
}
function removeClarifType(idx) {
  appState.clarifTargets.splice(idx, 1);
  saveState(appState, true);
  renderAdminView();
}
function addDocType() {
  const inp = document.getElementById('newDocTypeInput');
  const val = inp.value.trim();
  if (val) {
    appState.docCategories.push({ id: 'cat_' + Date.now(), name: val, subTypes: ['Standard'] });
    saveState(appState, true);
    renderAdminView();
    inp.value = '';
  }
}
function removeDocCategory(catIdx) {
```

```
      appState.docCategories.splice(catIdx, 1);
      saveState(appState, true);
      renderAdminView();
    }
    function addSubType(catIdx) {
      const inp = document.getElementById(`subInput_${catIdx}`);
      const val = inp.value.trim();
      if (val) {
        appState.docCategories[catIdx].subTypes.push(val);
        saveState(appState, true);
        renderAdminView();
      }
    }
    function removeSubType(catIdx, subIdx) {
      appState.docCategories[catIdx].subTypes.splice(subIdx, 1);
      saveState(appState, true);
      renderAdminView();
    }

    /* DÉMARRAGE */
    window.addEventListener('DOMContentLoaded', () => {
      initFirebase();
      if (appState.currentUser) {
        launchWorkspace();
      } else {
        showView('loginSelector');
        renderHeaderNav();
      }
    });
  </script>
</body>
</html>
```